

AHMAD MARAQA

AI Research Engineer | AI/ML · LLM Agents · Computer Vision · AI Automation

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[Portfolio](#) • [LinkedIn](#) • [GitHub](#)

PROFILE

AI Research Engineer working across AI/ML, large language models, computer vision and AI automation. I train and evaluate models on real, noisy data, build LLM systems on RAG, embeddings and tool-calling agents, and run the automation that connects them to databases, APIs and messaging platforms in production.

CORE SKILLS

AI & Machine Learning Classification · Regression · Multi-Label Classification · Ensemble Methods · Model Evaluation · Deep Learning · Neural Networks · CNNs · TensorFlow · Keras · Scikit-learn · Pandas · NumPy

LLMs & Agents RAG · Embeddings · Vector Databases (Pinecone, FAISS) · Prompt Engineering · Tool Calling · Structured Output · Local & Open-Source LLMs · Ollama · LangChain · LlamaIndex · OpenRouter · Groq

Computer Vision Object Detection · Image Classification · YOLOv8 · OpenCV · Ultralytics · CVAT · Data Annotation · Data Augmentation

Automation & APIs Python · n8n (incl. self-hosted) · MongoDB · WhatsApp Business Cloud API · Telegram Bot API · Google Workspace APIs · REST APIs · Webhooks · Webhook Security

Deployment & Ops Hostinger VPS · Cloudflare · Domains & DNS · Streamlit · GitHub Pages · Git · IoT · BLE · RTLS

PROFESSIONAL EXPERIENCE

AI Research & Integration Engineer — **Penguin Location Services** Jan 2025 – Present

- Research and implement AI/ML solutions for real-world applications, including machine learning models for IoT and RTLS environments.
- Develop LLM-based applications using RAG, embeddings, vector databases, prompt engineering and local/open-source LLMs via Ollama.
- Build AI-powered assistants and integrations connecting language models with external services and tools.

AI Automation & Agents — Independent — **Self-taught** 2025 – Present

- Develop and maintain production automation workflows using n8n, MongoDB, APIs, webhooks and external integrations — 30+ built to date.
- Integrate the WhatsApp Business Cloud API into automated workflows and AI-powered applications: approved templates, verified webhooks, rate-limit-aware batching, and timed maintenance jobs.
- Build tool-calling AI agents on the Telegram Bot API for task management, scheduling, document review and support with human escalation — integrated with Google Sheets, Calendar and Gmail, speech-to-text and text-to-speech, and RAG over Pinecone.
- Self-host and maintain n8n on a Hostinger VPS, including credentials, HTTPS, persistence, domains, DNS and Cloudflare.

AI Research Engineer Intern — **Penguin Location Services** Oct 2024 – Jan 2025

- Trained and evaluated KNN, Decision Tree and Random Forest models on BLE sensor data, and implemented a custom probabilistic model keeping prediction confidence in the output.

AI & ML Engineer Intern — **Leading Point Software** Jul 2024 – Sep 2024

- Built machine learning and computer vision solutions including image classification and object detection; hands-on with AutoML tools.

SELECTED PROJECTS

Jordanian Currency Detection & Counting — YOLOv8 detection over a dataset collected and augmented in-house, since no public one existed. Deployed as a web app. [\[Live\]](#) [\[Code\]](#)

Fire Detection from IoT Sensors — Binary classification on ~62k tabular readings across 13 features; seven models compared and surfaced side by side. [\[Live\]](#) [\[Code\]](#)

Rejecting Unknown Classes in Classification — Same CNN built twice, changing only the output layer and loss, so the model can decline inputs outside its training classes. [\[Notebook\]](#)

Dublejli — AI Video Dubbing System — Speech-to-text, translation and text-to-speech pipeline with speech-to-video synchronisation. Exhibited at SOFEX 2024 and NTP 2024.

EDUCATION

B.Sc. Computer Engineering — **University of Jordan** Oct 2020 – Sep 2024

GPA 3.77 / 4.00 — Excellent • Languages: Arabic (native), English (professional working proficiency)